

Лабораторная работа №1

Введение в Mininet

Шуваев С. А.

Российский университет дружбы народов, Москва, Россия

Информация

- Шуваев Сергей Александрович
- студент
- Российский университет дружбы народов
- 103224269@pfur.ru
- <https://Grinders060050.github.io/ru/>



Основной целью работы является развёртывание в системе виртуализации (например, в VirtualBox) mininet, знакомство с основными командами для работы с Mininet через командную строку и через графический интерфейс.

Выполнение лабораторной работы

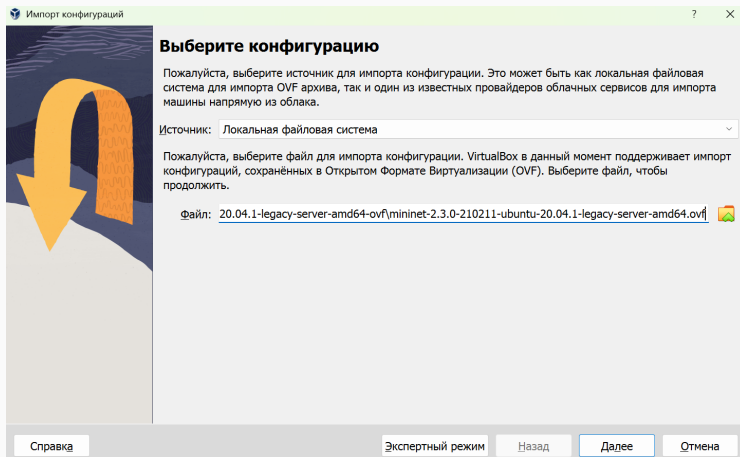


Figure 1: Импорт конфигураций

Выполнение лабораторной работы

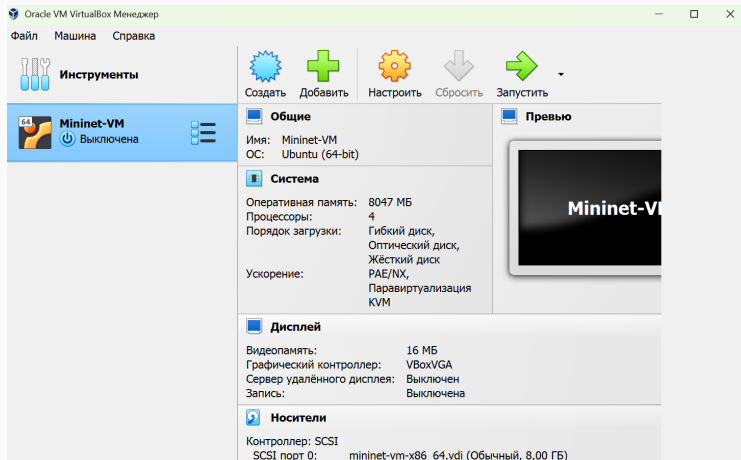


Figure 2: Параметры импорта

Выполнение лабораторной работы

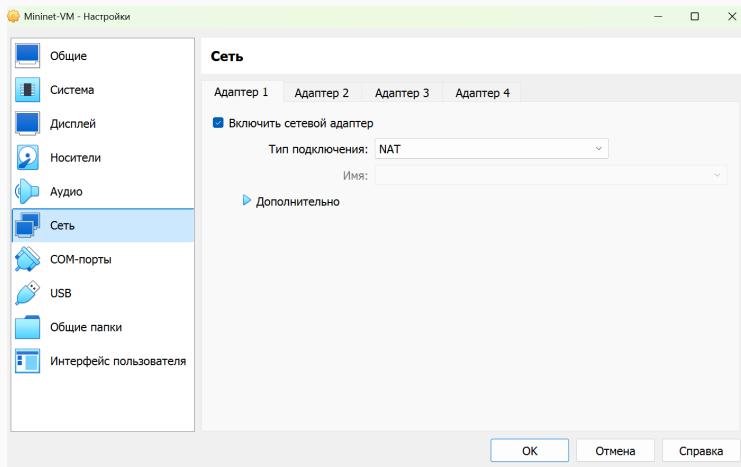


Figure 3: Настройка сети

Выполнение лабораторной работы

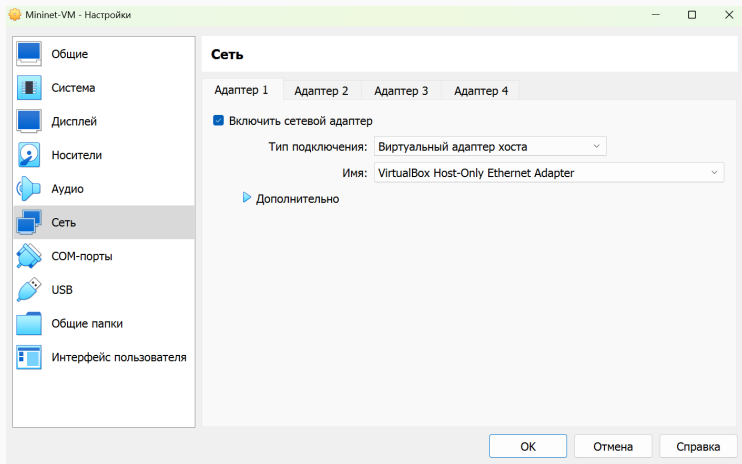


Figure 4: Настройка сети

Выполнение лабораторной работы

```
Ubuntu 20.04.1 LTS mininet-vm tty1
mininet-vm login: mininet
Password:
Welcome to Ubuntu 20.04.1 LTS (GNU/Linux 5.4.0-42-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

Last login: Wed Feb 10 21:03:31 PST 2021 on ttyS0
mininet@mininet-vm:~$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
    inet 192.168.56.102 netmask 255.255.255.0  broadcast 192.168.56.255
    ether 08:00:27:03:cc:2d  txqueuelen 1000  (Ethernet)
    RX packets 6  bytes 3058 (3.0 KB)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 2  bytes 684 (684.0 B)
    TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
    inet 127.0.0.1  netmask 255.0.0.0
    loop txqueuelen 1000  (Local Loopback)
    RX packets 128  bytes 9816 (9.8 KB)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 128  bytes 9816 (9.8 KB)
    TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0

mininet@mininet-vm:~$
```

Figure 5: Запуск mininet

Выполнение лабораторной работы

```
+ xauth list $DISPLAY
+
+ CategoryInfo          : ObjectNotFound: (xauth:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\WINDOWS\system32> xauth add mininet-vm/unix:10
xauth : Имя "xauth" не распознано как имя командлета, функции, файла сценария или выполняемой программы. Проверьте правильность написания имени, а также наличие и правильность пути, после чего повторите попытку.
строка:1 знак:1
+ xauth add mininet-vm/unix:10
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (xauth:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\WINDOWS\system32> ssh mininet@192.168.56.102
mininet@192.168.56.102's password:
Welcome to Ubuntu 20.04.1 LTS (GNU/Linux 5.4.0-42-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings

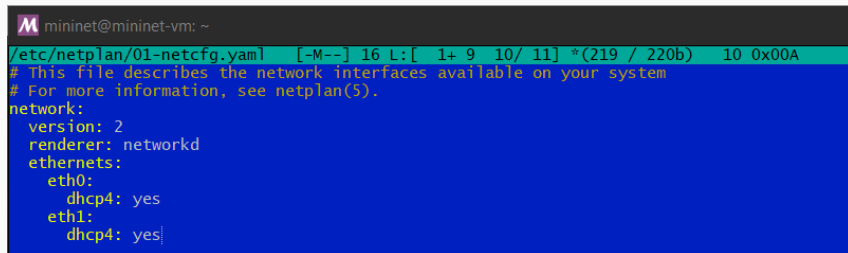
Last login: Sat Sep 13 06:47:37 2025 from 192.168.56.1
mininet@mininet-vm:~$ xauth add mininet-vm/unix:10 MIT-MAGIC-COOKIE-1 295acad8e35d17636924c5ab80e8462d
xauth: file /home/mininet/.Xauthority does not exist
mininet@mininet-vm:~$ xauth list $DISPLAY
mininet-vm/unix:10 MIT-MAGIC-COOKIE-1 295acad8e35d17636924c5ab80e8462d
mininet@mininet-vm:~$
```

Figure 6: Подключение к mininet через SSH

```
mininet@mininet-vm:~$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.56.102 netmask 255.255.255.0 broadcast 192.168.56.255
    ether 08:00:27:03:cc:24 txqueuelen 1000 (Ethernet)
    RX packets 342 bytes 79123 (79.1 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 161 bytes 22674 (22.6 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 2748 bytes 205684 (205.6 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 2748 bytes 205684 (205.6 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

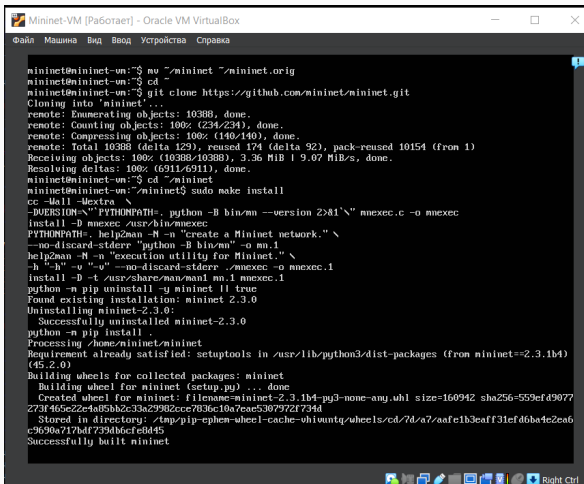
Figure 7: Просмотр IP-адресов машины



```
mininet@mininet-vm: ~  
/etc/netplan/01-netcfg.yaml [-M--] 16 L:[ 1+ 9 10/ 11] *(219 / 220b) 10 0x00A  
# This file describes the network interfaces available on your system  
# For more information, see netplan(5).  
network:  
  version: 2  
  renderer: networkd  
  ethernets:  
    eth0:  
      dhcp4: yes  
    eth1:  
      dhcp4: yes|
```

Figure 8: Файл /etc/netplan/01-netcfg.yaml

Выполнение лабораторной работы

The image shows a terminal window titled "Mininet-VM [Работает] - Oracle VM VirtualBox". The terminal output shows the following commands and their results:

```
mininet@mininet-vm:~$ mv ~/mininet ~/mininet.orig
mininet@mininet-vm:~$ cd -
mininet@mininet-vm:~$ git clone https://github.com/mininet/mininet.git
Cloning into 'mininet'...
remote: Enumerating objects: 10388, done.
remote: Counting objects: 100% (234/234), done.
remote: Compressing objects: 100% (140/140), done.
remote: Total 10388 (delta 129), reused 174 (delta 92), pack-reused 10154 (from 1)
Receiving objects: 100% (10388/10388), 3.36 MiB | 9.07 MiB/s, done.
Resolving deltas: 100% (6911/6911), done.
mininet@mininet-vm:~$ cd ~/mininet
mininet@mininet-vm:~/mininet$ sudo make install
cc -Wall -Wextra \
-DVERSION=\\"PYTHONPATH=. python -B bin/mn --version Z>81\\" mnexec.c -o mnexec
install -D mnexec /usr/bin/mnexec
PYTHONPATH=. help2man -N -n "create a Mininet network." \
--no-discard-stderr "python -B bin/mn" -o mn.1
help2man -N -n "execution utility for Mininet." \
-h "-h" -v "-v" --no-discard-stderr ./mnexec -o mnexec.1
install -D -t /usr/share/man/man1 mn.1 mnexec.1
python -n pip uninstall -y mininet || true
Found existing installation: mininet 2.3.0
Uninstalling mininet-2.3.0:
  Successfully uninstalled mininet-2.3.0
python -n pip install .
Processing /home/mininet/mininet
Requirement already satisfied: setuptools in /usr/lib/python3/dist-packages (from mininet==2.3.1b4) (45.2.0)
Building wheels for collected packages: mininet
  Building wheel for mininet (setup.py) ... done
  Created wheel for mininet: filename=mininet-2.3.1b4-py3-none-any.whl size=160942 sha256=559efd9077273f465e22e4a85bb2c33a29902cce7036e10a7eae5307972f734d
  Stored in directory: /tmp/pip-ephem-wheel-cache-vhiwvntq/wheels/cd/7d/a7/aaf1b3eaff31efd6ba4e2ea6c9690a717bdf739db6cf8d45
Successfully built mininet
```

Figure 9: Обновление Mininet

```
mininet@mininet-vm:~/mininet$ mn --version  
2.3.1b4  
mininet@mininet-vm:~/mininet$
```

Figure 10: Номер установленной версии mininet

Выполнение лабораторной работы

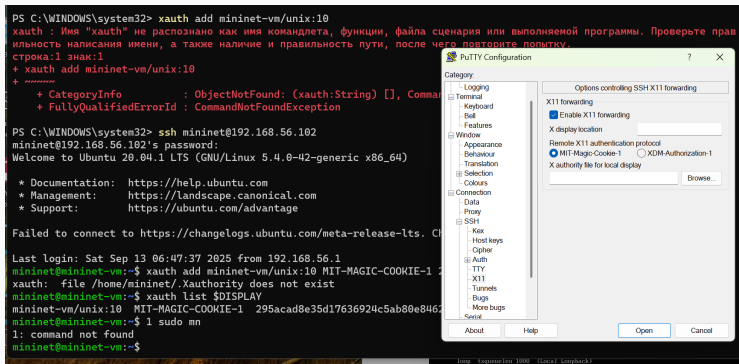


Figure 11: Настройка соединения X11 для суперпользователя

```
mininet@mininet-vm:~$ sudo dhclient eth1
Cannot find device "eth1"
mininet@mininet-vm:~$ sudo mn
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2
*** Adding switches:
s1
*** Adding links:
(h1, s1) (h2, s1)
*** Configuring hosts
h1 h2
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Starting CLI:
mininet> nodes
available nodes are:
c0 h1 h2 s1
mininet> net
h1 h1-eth0:s1-eth1
h2 h2-eth0:s1-eth2
s1 lo: s1-eth1:h1-eth0 s1-eth2:h2-eth0
c0
mininet>
```

Figure 12: Работа с Mininet с помощью командной строки

```
mininet> h1 ifconfig
h1-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.0.1 netmask 255.0.0.0 broadcast 10.255.255.255
    ether d2:06:bb:63:97:84 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

mininet>
```

Figure 13: Работа с Mininet с помощью командной строки

```
mininet> h2 ifconfig
h2-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.0.2 netmask 255.0.0.0 broadcast 10.255.255.255
    ether 8e:65:d6:ca:d4:38 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

Figure 14: Работа с Mininet с помощью командной строки

```
mininet> h1 ping 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=1.60 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.200 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.030 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.036 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.091 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.035 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.036 ms
^C
--- 10.0.0.2 ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6108ms
rtt min/avg/max/mdev = 0.030/0.290/1.603/0.538 ms
mininet> exit
```

Figure 15: Проверка связности хостов

В результате выполнения данной лабораторной работы я развёрнул mininet в системе виртуализации VirtualBox, а также ознакомился с основными командами для работы с Mininet через командную строку и через графический интерфейс.

1. Mininet [Электронный ресурс]. Mininet Project Contributors. URL: <http://mini-net.org/> (дата обращения: 11.12.2024).